

Complementary-Power Rigidity, a Gaussian Euler-Sum Identity, and Reflected Gamma Moments

Research continuation prepared for Vladimir Reshetnikov
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Abstract

We continue the ProveIt polylogarithm manuscript and its recent research packages. A classification theorem shows that two distinct positive integer pairs satisfying $r^a + r^b = 1$ can occur only for a root of the reciprocal plastic number, with pairs $(d, 5d)$ and $(2d, 3d)$. It gives the sharp bound of twelve ordered solutions of $x + y = 1$ inside the signed cyclic group $\{\pm r^k : k \in \mathbb{Z}\}$ and an effective degree bound for an exhaustive search. Classical functional equations then prove the manuscript's plastic dilogarithm and trilogarithm identities and give further identities without logarithmic terms.

We also prove the manuscript's weight-five Gaussian S_4 identity by an exact rational certificate built from double shuffle, distribution, and one explicit Möbius path duality. The certificate is independently replayable and does not use numerical integer-relation evidence as a proof. On the analytic side, we determine the exponential cancellation rate and first correction for alternating harmonic polylogarithms at proportional depth, and prove meromorphic continuation, finite-order asymptotics, late-coefficient asymptotics, and an explicit growing-order remainder bound for reflected log-gamma moments. A finite identity among their residues and a formal proof of a recorded log-sine coefficient conjecture complete the development. We state the exact scope of each result, document proposed manuscript corrections, and formulate further questions. No numerical period independence or general maximal ladder weight is claimed.

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1 Scope, prior work, and the proved contributions

The source manuscript is *Polylogarithms and their Arithmetic Bridges*, in the directory `Analysis/Polylogarithms/docs/manuscript` of ProveIt [1]. We fixed the inspected revision at

d00eba30c1ce73ee5cfbeb50bde9095246b9c25c.

That revision contains the consolidated manuscript, its earlier reports, and four relevant incoming archives. The present article is an additional research contribution prepared for integration; it is not a replacement for the whole manuscript or a certification of every historical claim.

The immediately preceding reports have already proved Gaussian double parity formulas, the weight-five shuffle relation, several one-index-two multiple-polylogarithm reductions, and the cubic and quartic complementary-power trilogarithm examples. They also reduce the four same-point weight-five Gaussian coordinates to at most two modulo specified single-value products. We use those facts where appropriate and do not count their rederivation as a new contribution. In particular, the first follow-up problem in the exact-reductions package is the specific S_4 relation proved below; its fifth asks for a classification of complementary-power redundancies [2].

1.1 A precise contribution ledger

Result	Mathematical content and proof status
Complementary-power rigidity	A complete theorem for two positive integer exponent pairs at the same real base, obtained from Ljunggren's trinomial theorem and a root-rotation argument. It includes equal exponents and arbitrary common divisors.
Signed cyclic seeds	Exact cardinalities 0, 3, 6, 12, with the case twelve characterized; a bound $b \leq \lfloor 5D/3 \rfloor$ for bases of degree at most D . These are statements about pure powers.
Plastic ladders	Analytic proofs of two specific manuscript identities, plus explicit dilogarithm and trilogarithm identities with no logarithmic terms. Classical functional equations are the inputs.
Gaussian S_4	A proved identity with a finite exact certificate. The new ingredient relative to the preceding attempts is an explicit Möbius duality relation.
Proportional harmonic depth	An explicit positive, decreasing, convex exponential rate and a computable first correction, with an all-orders expansion uniform on compact parameter sets.
Reflected gamma moments	Exact poles and residue polynomials, shifted factorial growth, late-coefficient asymptotics, and an explicit truncation bound.
Residue and log-sine identities	An exact finite reflection identity and an unconditional formal polynomial version of a coefficient conjecture recorded in a 2010 author preprint.

The classification gives a particularly useful change in how one can search. A box of polynomial coefficients is not exhaustive, even at fixed degree. In contrast, the exponent bound proved here supplies an exhaustive finite search for the restricted equation $r^a + r^b = 1$. Equally importantly, it

identifies exactly what this search cannot settle: multiplicative relations involving products of many factors $1 - r^j$, additional units, or more general polylogarithm arguments are outside its scope.

The S_4 proof has a different character. It is a finite calculation with convergent iterated-integral values, using explicitly justified single-divergence relations as well as convergent identities. Its exact certificate can be checked by rational arithmetic. The mathematical conclusion is an equality of the corresponding periods. It is not a proof that a computed quotient of all supplied relations is the full space of period relations, nor does it prove that any remaining coordinate is irreducible.

1.2 Classical inputs and attribution

Ljunggren’s irreducibility theorem is used in the precise form recorded by Boyd, Martin, and Thom [6, Lemma 4.1]; Tverberg’s proof is another primary reference [5]. Rogers’ pentagon and Kummer’s trilogarithm equation are classical; the modified trilogarithm normalization and its functional equations are taken from Zagier [8]. The shuffle and stuffle laws and their regularization belong to the established theory of multiple polylogarithms [13, 15]. The general existence of parity reductions is therefore not being claimed here as new.

The analytic calculations use standard Mellin continuation, Lagrange inversion, and Laplace asymptotics. What is derived is their explicit application to the stated families, including normalization constants, finite coefficient identities, and error terms. Mixed reflected log-gamma moments were already introduced by Amdeberhan and coauthors [16]; neither their definition nor their reflection symmetry is a new observation. Their large-exponent and late-coefficient behavior is the subject of the later sections.

This article establishes the statements it proves without asserting first discovery in the entire literature. Exact priority for equivalent normalizations of special-value identities is a separate bibliographic question. The useful distinction is between proved results, numerical checks of those results, and questions that remain open.

1.3 Conventions

For positive integers $s_1, \dots, s_d$, our multiple polylogarithm uses decreasing indices:

$$\mathrm{Li}_{s_1, \dots, s_d}(z_1, \dots, z_d) = \sum_{n_1 > \dots > n_d \geq 1} \frac{z_1^{n_1} \dots z_d^{n_d}}{n_1^{s_1} \dots n_d^{s_d}}. \quad (1.1)$$

Boundary values are assigned by the ordinary convergent series or by the explicitly stated regularization. The corresponding hyperlogarithm uses $dt/(t-a)$ and an outer-first word, so its integration simplex is $0 < t_N < \dots < t_1 < 1$. The word letters associated to (1.1) are reciprocal *prefix* products. These choices are essential to the signs in the Gaussian certificate.

All logarithms in the real algebraic and gamma sections are real logarithms of positive numbers. Complex arguments in the Gaussian section are connected to their convergent series, fixing the branches. We write $G = \beta(2)$ for Catalan’s constant and $\beta(s) = \sum_{j \geq 0} (-1)^j / (2j+1)^s$. The letter h denotes harmonic depth in Section 5; the fixed reflected exponent in the gamma section is m . Repeated use of an auxiliary symbol in a later, explicitly defined family does not identify the corresponding constants across sections.

2 Rigidity of complementary powers

The earlier complementary-power ladder starts from a relation $r^a + r^b = 1$, where $0 < r < 1$ and a, b are positive integers. That construction raises a concrete completeness question: can one base have many different pairs of complementary powers? We settle this question for pure powers, including arbitrary common divisors in the exponents. The decisive arithmetic input is a classical irreducibility theorem for trinomials. Our conclusions concern this precisely specified class of seeds; they do not classify general multiplicative relations among $1 - r^k$, or bound the maximum weight of polylogarithm ladders.

For $0 < r < 1$, define

$$\mathcal{C}(r) = \{(a, b) \in \mathbb{Z}_{>0}^2 : a \leq b, r^a + r^b = 1\}.$$

Let ω be the unique root in $(0, 1)$ of $X^3 + X^2 - 1$. Thus ω is the reciprocal of the plastic number.

Theorem 2.1 (Complete classification of collisions). *For every real $0 < r < 1$, the set $\mathcal{C}(r)$ has at most two elements. It has two elements if and only if $r^d = \omega$ for some positive integer d , in which case*

$$\mathcal{C}(r) = \{(d, 5d), (2d, 3d)\}.$$

If $\mathcal{C}(r)$ contains (a, a) , then $\mathcal{C}(r) = \{(a, a)\}$ and $r = 2^{-1/a}$.

The theorem supplies an exact answer to the seed-classification question posed after the earlier cubic and quartic ladder certificates. It is proved below as a consequence of the Ljunggren–Tverberg theorem, with all additional arguments included. We make no claim that this consequence is globally new in the literature. The related theorem of Aarts, Fokkink, and Kruijtzter classifies the two morphic numbers, by requiring both $p + 1$ and $p - 1$ to be powers of p ; their proof also uses trinomial irreducibility [7]. Our formulation permits arbitrary positive exponent pairs and explicitly handles power substitutions.

2.1 The irreducibility input and the entire cyclotomic factor

For $1 \leq a < b$, put $d = \gcd(a, b)$, $A = a/d$, $B = b/d$, and

$$F_{a,b}(X) = X^b + X^a - 1.$$

The following precise specialization is the arithmetic input.

Theorem 2.2 (Ljunggren–Tverberg, specialized). *The polynomial $F_{a,b}$ has exactly one irreducible factor over $\mathbb{Q}$ whose roots are not roots of unity. Its other factor is*

$$C_{a,b}(X) = \begin{cases} X^{2d} - X^d + 1, & (A, B) \equiv (1, 5) \text{ or } (5, 1) \pmod{6}, \\ 1, & \text{otherwise.} \end{cases}$$

In particular $H_{a,b} = F_{a,b}/C_{a,b}$ is irreducible, monic, and has constant term -1 . Moreover $H_{a,b}(X) = H_{A,B}(X^d)$.

This is the trinomial result of Ljunggren and Tverberg [4, 5]; a modern explicit formulation, including the assertion that $H_{A,B}(X^d)$ remains irreducible, is Boyd–Martin–Thom, Lemma 4.1(a),(b) [6]. Only the trinomial theorem is used. Its separate quadrinomial analogue has a more delicate history and plays no role here.

For clarity, the displayed cyclotomic factor can be obtained directly, without relying on a table of sign cases. Suppose $|z| = 1$ and $z^a + z^b = 1$. The two unit vectors must be the two numbers $e^{i\pi/3}$ and $e^{-i\pi/3}$, in either order. If $w = z^d$, then w^A and w^B are these primitive sixth roots. A Bezout identity for A, B implies $w^6 = 1$, and w^A of order six then implies that w itself has order six. Consequently A, B have opposite residues 1, 5 modulo six. Conversely those residues make every root of $X^{2d} - X^d + 1$ a root of $F_{a,b}$. These roots are simple: a common zero of $F_{a,b}$ and its derivative on the unit circle would give $a|z^a| = b|z^b|$, hence $a = b$, a contradiction. Thus the formula for $C_{a,b}$ accounts for the *entire* cyclotomic factor. The remaining assertion, irreducibility of $H_{a,b}$, is the cited theorem.

2.2 A conjugate-rotation invariant

Lemma 2.3 (The common divisor is intrinsic). *Suppose $r^a + r^b = r^c + r^e = 1$, where $0 < r < 1$, $1 \leq a < b$, and $1 \leq c < e$. Then*

$$\gcd(a, b) = \gcd(c, e).$$

Proof. Set $d = \gcd(a, b)$. Since $0 < r < 1$, it is not a root of unity, so its minimal polynomial is $H_{a,b}$. Theorem 2.2 shows that this polynomial is a polynomial in X^d . If ζ is a primitive d th root of unity, $r\zeta$ is therefore a conjugate of r . The second equation also holds at this conjugate:

$$r^c \zeta^c + r^e \zeta^e = 1.$$

Taking absolute values gives equality in the triangle inequality, because $r^c + r^e = 1$. Both summands must have the argument of their positive real sum. Thus $\zeta^c = \zeta^e = 1$, proving $d \mid \gcd(c, e)$. Interchanging the pairs proves the reverse divisibility. $\square$

This step prevents a gap that would arise from normalizing the two pairs independently. The cyclotomic factor can have degree $2d$, so a degree comparison before proving that the common divisors agree would not suffice.

2.3 The primitive coefficient comparison

Proof of Theorem 2.1. First suppose an equal pair occurs. Then $r^a = 1/2$, so r cannot be an algebraic integer: otherwise the rational number $r^a = 1/2$ would be an algebraic integer. Any unequal pair would make r a root of the monic integer polynomial $F_{c,e}$, a contradiction. There is only one equal pair because the powers of r strictly decrease.

Now suppose two distinct unequal pairs occur. By Lemma 2.3 they have a common gcd d . Replace r by r^d and divide all four exponents by d . Both pairs are now primitive. Their trinomials have the same irreducible noncyclotomic factor, and their cyclotomic factors belong to $\{1, \Phi_6\}$, where $\Phi_6 = X^2 - X + 1$. If the two cyclotomic factors agree, the trinomials agree, contrary to distinctness. After interchanging them if necessary, there must therefore be a polynomial identity

$$(X^2 - X + 1)(X^n + X^m - 1) = X^{n+2} + X^k - 1, \quad 1 \leq m < n, \quad 1 \leq k < n + 2. \quad (2.1)$$

The coefficient of X^{n+1} on the left is -1 unless $m = n - 1$, in which case it is zero. The right-hand coefficient is zero or one. Hence $m = n - 1$, and the identity reduces to

$$X^{n+2} + X^{n-1} - X^2 + X - 1 = X^{n+2} + X^k - 1.$$

For $n = 2$ the remaining polynomial is $2X - X^2$, which is not a monomial. For $n > 3$ its coefficient at X^2 is -1 . Thus $n = 3$, which gives $m = 2$ and $k = 1$. The common factor is $X^3 + X^2 - 1$, and the two normalized pairs are $(1, 5)$ and $(2, 3)$. Conversely,

$$X^5 + X - 1 = (X^2 - X + 1)(X^3 + X^2 - 1)$$

shows that these two pairs do occur at ω . Finally, a third pair would have to be the same exceptional companion of either existing pair, so no third pair occurs. $\square$

2.4 All signed pure-power unit equations

The functional equations for polylogarithms use signs and inverses as well as positive powers. The preceding classification automatically controls all such complementary arguments.

Corollary 2.4 (Sharp signed-unit count). *Let $\Gamma_r = \{\pm r^k : k \in \mathbb{Z}\}$ and*

$$\mathcal{U}(r) = \{x \in \Gamma_r : 1 - x \in \Gamma_r\}.$$

Then $|\mathcal{U}(r)|$ is one of $0, 3, 6, 12$. For every unequal pair $(a, b) \in \mathcal{C}(r)$, its contribution is the six-element set

$$\{r^a, r^b, r^{-a}, r^{-b}, -r^{b-a}, -r^{a-b}\}. \quad (2.2)$$

These sets are disjoint for different unordered pairs. An equal pair contributes precisely $\{1/2, 2, -1\}$. The case $|\mathcal{U}(r)| = 12$ occurs exactly when $r^d = \omega$ for some positive integer d .

Proof. The transformations $x \mapsto 1 - x$ and $x \mapsto 1/x$ preserve $\mathcal{U}(r)$. For example, $1 - 1/x = -(1 - x)/x \in \Gamma_r$ whenever $x, 1 - x \in \Gamma_r$. Their orbit consists of

$$x, \quad 1 - x, \quad x^{-1}, \quad (1 - x)^{-1}, \quad x/(x - 1), \quad (x - 1)/x.$$

Every real orbit has a representative in $(0, 1)$: use $1/x$ if $x > 1$ and $1/(1 - x)$ if $x < 0$. Such a representative and its complement are r^a, r^b with $a, b > 0$. The two members of a nondegenerate real orbit that lie in $(0, 1)$ are exactly x and $1 - x$, so its unordered positive pair is unique. Substituting $x = r^a$, $1 - x = r^b$ gives (2.2). All six entries are distinct unless $a = b$, when they reduce to the stated three values. The cardinalities now follow from Theorem 2.1. $\square$

All four possibilities occur: $r = 1/3, 1/2$, the reciprocal golden ratio, and ω give respectively $0, 3, 6, 12$. The conclusion classifies *pure signed-power complements*. It does not say that there are at most twelve useful polylogarithm arguments: products of units $1 - r^k$ can introduce many more arguments.

Corollary 2.5 (Rational exponents). *The same collision classification holds for positive rational exponents, with the exceptional common scale d allowed to be a positive rational number. In particular there are at most two unordered positive rational pairs, and*

$$\#\{x \in \{\pm r^q : q \in \mathbb{Q}\} : 1 - x \in \{\pm r^q : q \in \mathbb{Q}\}\} \leq 12.$$

Proof. Given two pairs, choose a common denominator N and apply Theorem 2.1 to $r^{1/N}$. A third rational pair could be included in the same denominator and would contradict the integer theorem. The orbit proof is unchanged. $\square$

2.5 A provably complete bounded-degree search

Corollary 2.6 (Degree and an exhaustive exponent bound). *If $r^a + r^b = 1$ with $1 \leq a < b$, then*

$$[\mathbb{Q}(r) : \mathbb{Q}] = \begin{cases} b - 2d, & (a/d, b/d) \equiv (1, 5) \text{ or } (5, 1) \pmod{6}, \\ b, & \text{otherwise,} \end{cases} \quad d = \gcd(a, b).$$

Consequently all such bases of degree at most D are found by enumerating

$$1 \leq a < b \leq \left\lfloor \frac{5D}{3} \right\rfloor$$

and retaining those whose displayed degree is at most D . For primitive pairs it suffices to enumerate $b \leq D + 2$. The only repeated bases in either enumeration are the classified plastic pairs. Equal pairs add the separate bases $2^{-1/a}$, with $a \leq D$.

Proof. The degree formula is Theorem 2.2. In the exceptional case $b/d \geq 5$, so $b - 2d \geq 3d$. If the degree is at most D , then $d \leq D/3$ and $b = (b - 2d) + 2d \leq 5D/3$. The nonexceptional case is immediate. For primitive pairs the possible degree loss is only two. The equal-pair polynomial $2X^a - 1$ has degree a and is irreducible: its reciprocal is, up to a nonzero scalar, $X^a - 2$, which is Eisenstein at 2. $\square$

This bound is sharp for infinitely many D : the pair $(d, 5d)$ has degree $3d$ and largest exponent $5d$. Unlike a search over bounded polynomial coefficients, the algorithm has an explicit completeness theorem for its stated class. No numerical equality test is required. A base is stored by the exact irreducible polynomial $H_{a,b}$ together with its unique root in $(0, 1)$, and duplicate detection is exact polynomial comparison.

For illustration, the complete unequal-pair list through degree four is

degree	complement pair(s)	minimal polynomial
2	(1, 2)	$X^2 + X - 1$
3	(1, 3)	$X^3 + X - 1$
3	(1, 5), (2, 3)	$X^3 + X^2 - 1$
4	(1, 4)	$X^4 + X - 1$
4	(2, 4)	$X^4 + X^2 - 1$
4	(3, 4)	$X^4 + X^3 - 1$

The degree-four entry with pair $(2, 4)$ is exactly a square root of the reciprocal golden ratio. Its presence illustrates why normalization of exponents is necessary when counting bases rather than primitive seed types.

The exact enumeration in the accompanying certificate gives:

D	1	2	3	4	5	6	7	8	9	10	11	12
unequal-pair bases	0	1	3	6	10	15	20	27	37	46	56	67
primitive bases	0	1	3	5	9	11	16	20	28	32	42	46
all bases	1	3	6	10	15	21	27	35	46	56	67	79

Each row counts bases of degree at most D . A primitive base has an unequal complementary pair of gcd one; the intrinsic-gcd lemma makes this definition independent of the choice of pair. The last row also includes the D equal-pair bases $2^{-1/a}$, $1 \leq a \leq D$.

2.6 An exact counting formula

The classification also counts bases without constructing or factoring their minimal polynomials. For $n \geq 1$ define

$$e(n) = \#\{a : 1 \leq a < n, \gcd(a, n) = 1, (a, n) \equiv (1, 5) \text{ or } (5, 1) \pmod{6}\}.$$

In particular $e(n) = 0$ unless $n \equiv 1$ or $5 \pmod{6}$, and $e(1) = 0$.

Corollary 2.7 (Counting all bases by algebraic degree). *Let $N(D)$ be the number of real $0 < r < 1$ with $[\mathbb{Q}(r) : \mathbb{Q}] \leq D$ and $\mathcal{C}(r) \neq \emptyset$. For every positive integer D ,*

$$\begin{aligned} N(D) = & \frac{D(D+1)}{2} - \left\lfloor \frac{D}{3} \right\rfloor \\ & + \sum_{d=1}^{\lfloor D/3 \rfloor} \left\{ e\left(\left\lfloor \frac{D}{d} \right\rfloor + 1\right) + e\left(\left\lfloor \frac{D}{d} \right\rfloor + 2\right) \right\}. \end{aligned} \quad (2.3)$$

In particular,

$$N(D) = \frac{1}{2}D^2 + O(D \log D), \quad D \rightarrow \infty.$$

Proof. There are $D(D-1)/2$ unequal pairs with largest exponent at most D , and their associated bases all have degree at most D . A pair with largest exponent greater than D can still have degree at most D only in the exceptional cyclotomic case. Write that pair (dA, dB) , where $\gcd(A, B) = 1$ and the residues are opposite 1, 5 modulo six. The two inequalities are

$$dB > D, \quad d(B-2) \leq D.$$

Equivalently, $B \in \{\lfloor D/d \rfloor + 1, \lfloor D/d \rfloor + 2\}$. Since the exceptional normalized degree $B-2$ is at least three, only $d \leq D/3$ can occur. This gives the sum in (2.3). Every base counted twice is $\omega^{1/d}$, of degree $3d$; there are exactly $\lfloor D/3 \rfloor$ such duplicates. Finally the D equal-pair bases are distinct from all unequal-pair bases and from each other. This proves the exact formula.

For the asymptotic statement, use $0 \leq e(n) \leq n$ to bound the sum by

$$\sum_{d \leq D/3} \left(2 \left\lfloor \frac{D}{d} \right\rfloor + 3 \right) \leq 2D \sum_{d \leq D/3} \frac{1}{d} + D.$$

The harmonic sum is $O(\log D)$, proving the claim. $\square$

For an alternative exact implementation, when $n > 1$ is coprime to six, Mobius inversion gives

$$e(n) = \sum_{d|n} \mu(d) \left\lfloor \frac{n/d+1}{6} \right\rfloor. \quad (2.4)$$

Indeed, after writing $a = dj$, the condition $a \equiv -n \pmod{6}$ becomes $j \equiv -(n/d) \pmod{6}$ because every unit modulo six is its own inverse. For t coprime to six, the number of integers $1 \leq j < t$ in the residue class $-t$ is $\lfloor (t+1)/6 \rfloor$. Summing with $\mu(d)$ imposes $\gcd(a, n) = 1$.

3 Exact proofs and log-free ladders at the plastic base

The exceptional base in Theorem 2.1 has two complementary pairs. This additional structure produces short proofs of two previously numerically supported formulas in the manuscript, as well as log-free identities for $\zeta(2)$ and $\zeta(3)$.

Throughout this section, $0 < q < 1$, $q^2 + q^3 = 1$, and $\lambda = \log q$. The factorization in the preceding proof gives

$$1 - q = q^5, \quad 1 - q^2 = q^3, \quad 1 - q^3 = q^2, \quad 1 - q^5 = q. \quad (3.1)$$

All logarithms used in the final identities are real.

3.1 The dilogarithm formula: one Rogers pentagon

Write

$$R(x) = \text{Li}_2(x) + \frac{1}{2} \log x \log(1 - x), \quad 0 < x < 1.$$

The Rogers reflection and pentagon equations are

$$R(x) + R(1 - x) = \zeta(2), \quad (3.2)$$

$$R(x) + R(y) = R(xy) + R\left(\frac{x(1 - y)}{1 - xy}\right) + R\left(\frac{y(1 - x)}{1 - xy}\right). \quad (3.3)$$

They hold for $0 < x, y < 1$ and follow from the classical dilogarithm functional equations; alternatively one can differentiate in x and evaluate at $x = 0$. This real-domain formulation avoids branch choices [8].

Theorem 3.1 (The manuscript's plastic dilogarithm formula).

$$\frac{1}{2} \text{Li}_2(q^2) - \text{Li}_2(q^5) = \lambda^2. \quad (3.4)$$

Proof. Set $R_k = R(q^k)$. At $(x, y) = (q, q^2)$ the three pentagon arguments are

$$xy = q^3, \quad \frac{x(1 - y)}{1 - xy} = q^2, \quad \frac{y(1 - x)}{1 - xy} = q^5.$$

Hence $R_1 + R_2 = R_3 + R_2 + R_5$, or $R_1 = R_3 + R_5$. Reflection gives $R_1 + R_5 = R_2 + R_3 = \zeta(2)$, so $R_2 = 2R_5$. Expanding the logarithmic corrections by (3.1) gives

$$\text{Li}_2(q^2) + 3\lambda^2 = 2\text{Li}_2(q^5) + 5\lambda^2,$$

as required. □

3.2 The trilogarithm formula: one Kummer specialization

For real nonzero $z \neq 1$, define the single-valued trilogarithm

$$P(z) = \text{Re Li}_3(z) - \log |z| \text{Re Li}_2(z) - \frac{1}{3} \log^2 |z| \log |1 - z|,$$

with the continuous value $P(1) = \zeta(3)$. The inversion and duplication equations are

$$P(z^{-1}) = P(z), \quad P(z) + P(-z) = \frac{1}{4} P(z^2).$$

Kummer's nine-term equation, in this single-valued real form, is

$$\begin{aligned} & 2P(x) + 2P(y) + 2P\left(\frac{x(1-y)}{x-1}\right) + 2P\left(\frac{y(1-x)}{y-1}\right) \\ & + 2P\left(\frac{1-x}{1-y}\right) + 2P\left(\frac{x(1-y)}{y(1-x)}\right) - P(xy) - P(x/y) \\ & - P\left(\frac{x(1-y)^2}{y(1-x)^2}\right) = 2\zeta(3). \end{aligned} \quad (3.5)$$

We use the form recorded by Zagier, Section 7, Example 3, equation (7) [8].

The single-valued Landen equation is

$$P(x) + P(1-x) + P(1-x^{-1}) = \zeta(3), \quad 0 < x < 1.$$

Its ordinary real counterpart is

$$\begin{aligned} \text{Li}_3(x) + \text{Li}_3(1-x) + \text{Li}_3(1-x^{-1}) &= \zeta(3) + \zeta(2) \log x \\ &\quad - \frac{1}{2} \log^2 x \log(1-x) + \frac{1}{6} \log^3 x. \end{aligned} \quad (3.6)$$

One obtains it by differentiation using dilogarithm reflection and Landen, with the integration constant determined at $x = 1$. See also Nakamura-Shiraishi [9].

Theorem 3.2 (The manuscript's plastic trilogarithm formula).

$$-\text{Li}_3(q) + \frac{3}{2} \text{Li}_3(q^2) + \frac{1}{3} \text{Li}_3(q^3) - \text{Li}_3(q^5) = \lambda^3. \quad (3.7)$$

Proof. Put $P_k = P(q^k)$. At $(x, y) = (q, q^2)$, the nine arguments in (3.5), in displayed order, are

$$q, \quad q^2, \quad -q^{-1}, \quad -q^4, \quad q^2, \quad q^{-3}, \quad q^3, \quad q^{-1}, \quad q^{-5}.$$

Using inversion and duplication, the equation becomes

$$\mathbf{K} = -P_1 + \frac{9}{2}P_2 + P_3 - 2P_4 - P_5 + \frac{1}{2}P_8 = 2\zeta(3).$$

Landen at $x = q$, with $1 - q = q^5$, similarly gives

$$\mathbf{A} = P_1 + P_5 - P_4 + \frac{1}{4}P_8 = \zeta(3).$$

Consequently the exact row operation $(\mathbf{K} - 2\mathbf{A})/3$ proves

$$-P_1 + \frac{3}{2}P_2 + \frac{1}{3}P_3 - P_5 = 0. \quad (3.8)$$

It remains to convert this to ordinary trilogarithms. Denote the left side of (3.7) by D , and put

$$S = -\text{Li}_2(q) + 3\text{Li}_2(q^2) + \text{Li}_2(q^3) - 5\text{Li}_2(q^5).$$

The two ordinary reflection equations and Theorem 3.1 give

$$\begin{aligned} \text{Li}_2(q) + \text{Li}_2(q^5) &= \zeta(2) - 5\lambda^2, \\ \text{Li}_2(q^2) + \text{Li}_2(q^3) &= \zeta(2) - 6\lambda^2, \\ \frac{1}{2} \text{Li}_2(q^2) - \text{Li}_2(q^5) &= \lambda^2. \end{aligned}$$

Taking minus the first row, plus the second, plus four times the third yields $S = 3\lambda^2$. The remaining logarithmic coefficient in the definition of P is

$$T = \{-1 \cdot 1^2 \cdot 5 + (3/2) \cdot 2^2 \cdot 3 + (1/3) \cdot 3^2 \cdot 2 - 1 \cdot 5^2 \cdot 1\}\lambda = -6\lambda.$$

Thus the left side of (3.8) is

$$D - \lambda S - \frac{\lambda^2}{3}T = D - 3\lambda^3 + 2\lambda^3 = D - \lambda^3.$$

This proves (3.7). □

The proof promotes the two displayed manuscript formulas from numerical support to exact consequences of classical functional equations. It does not rely on pseudointegration at an isolated algebraic base.

3.3 Two log-free identities

Theorem 3.3 (Log-free plastic ladders). *At the same base q , the following identities hold:*

$$6 \operatorname{Li}_2(q) - 5 \operatorname{Li}_2(q^2) - 5 \operatorname{Li}_2(q^3) + 6 \operatorname{Li}_2(q^5) = \zeta(2), \quad (3.9)$$

$$12 \operatorname{Li}_3(q) - 5 \operatorname{Li}_3(q^2) - 4 \operatorname{Li}_3(q^3) - 8 \operatorname{Li}_3(q^4) + 8 \operatorname{Li}_3(q^5) + 2 \operatorname{Li}_3(q^8) = 4\zeta(3). \quad (3.10)$$

Proof. Six times the first ordinary reflection equation minus five times the second proves (3.9). For the trilogarithm, write $L_k = \operatorname{Li}_3(q^k)$. Equation (3.6) at $x = q$ and $x = q^2$, with duplication to eliminate negative arguments, gives

$$\begin{aligned} A &= L_1 + L_5 - L_4 + \frac{1}{4}L_8 = \zeta(3) + \zeta(2)\lambda - \frac{7}{3}\lambda^3, \\ B &= \frac{5}{4}L_2 + L_3 - L_1 = \zeta(3) + 2\zeta(2)\lambda - \frac{14}{3}\lambda^3. \end{aligned}$$

The logarithmic tails are proportional. Therefore $4(2A - B) = 4\zeta(3)$, exactly the claimed six-argument identity. □

These formulas are explicit additional identities for the package. Their derivations and the classical origins of the functional equations are given in full; no global priority claim is made. They also give exponentially convergent algebraic series, for example

$$\zeta(3) = \sum_{n=1}^{\infty} \frac{3q^n - \frac{5}{4}q^{2n} - q^{3n} - 2q^{4n} + 2q^{5n} + \frac{1}{2}q^{8n}}{n^3}. \quad (3.11)$$

Absolute convergence justifies expansion of the polylogarithms. If this series is truncated after N terms, its absolute tail is at most

$$\frac{39}{4} \frac{q^{N+1}}{(N+1)^3(1-q)},$$

obtained by summing the absolute coefficients and using $q^{kn} \leq q^n$ for $k \geq 1$. The constant $39/4$ is deliberately simple; the sharper sum of the six individual geometric tails can also be used.

3.4 Exact certificates and their scope

The accompanying script checks the polynomial identities and all three Rogers and nine Kummer argument reductions in $\mathbb{Q}[X]/(X^3 + X^2 - 1)$. It constructs the Kummer row from the signed exponent list and performs the displayed rational row operations. Separately, it enumerates a bounded collection of trinomials, checks the predicted cyclotomic factors and irreducible quotients with exact polynomial arithmetic, and groups exact minimal polynomials to verify the expected collision table. The finite enumeration is a reproducible consistency check; the unbounded completeness result is Theorem 2.1.

4 The mixed Gaussian weight-five conjecture is a theorem

Put $G = \beta(2)$ and

$$g_{a,b} = \text{Im Li}_{a,b}(i, 1), \quad S_p = \sum_{n=0}^{\infty} \frac{(-1)^n H_n}{(2n+1)^p}.$$

The manuscript's equation `gauss:eq:S4-closed` was explicitly classified as a numerically supported identity. We prove precisely that identity. The contribution is a reduction of its mixed Gaussian term to three specified single-color doubles and elementary products. It is not a claim that the remaining doubles are independent or irreducible, nor a claim that S_4 previously lacked every representation in special functions. Indeed the 2021 discussion by Hintze and pisco already records the representation by two colored double values and relates it to hypergeometric evaluations [10, 11].

Theorem 4.1 (Certified mixed Gaussian reduction). *With the series convention $\text{Li}_{a,b}(x, y) = \sum_{n>m \geq 1} x^n y^m / (n^a m^b)$,*

$$S_4 = \frac{4g_{4,1} - 3g_{3,2} - 9g_{2,3}}{7} + \frac{\pi^5}{224} - \frac{27}{224} G\zeta(3) - 2\beta(4) \log 2. \quad (4.1)$$

Equivalently,

$$\text{Im Li}_{4,1}(i, -1) = -\frac{3g_{4,1} + 3g_{3,2} + 9g_{2,3}}{7} + \frac{\pi^5}{224} - \frac{27}{224} G\zeta(3) - 2\beta(4) \log 2. \quad (4.2)$$

The proof consists of convergent changes of variables, standard shuffle/stuffle identities with the explicitly described single-divergence regularization, and a finite rational linear combination supplied as an independently replayable certificate.

4.1 The arithmetic translation

For $n = 2k + 1$,

$$H_k = \sum_{m=1}^{n-1} \frac{1 + (-1)^m}{m}.$$

Taking imaginary parts removes even n , while $\text{Im}(i^{2k+1}) = (-1)^k$. Consequently, for $p \geq 2$,

$$S_p = \text{Im}(\text{Li}_{p,1}(i, 1) + \text{Li}_{p,1}(i, -1)). \quad (4.3)$$

The interchange is justified by the majorant $\sum_{n \geq 2} H_{n-1}/n^p < \infty$. Thus (4.1) and (4.2) are equivalent.

4.2 A convergent octahedral change of variables

Write $\omega_a = dt/(t - a)$ and

$$\mathcal{I}(a_1 \cdots a_w) = \int_{0 < t_w < \cdots < t_1 < 1} \omega_{a_1}(t_1) \cdots \omega_{a_w}(t_w),$$

extended multilinearly in the letters. A convergent word has first letter different from 1 and last letter different from 0. For colors $c_j \in \mathbb{Z}/4\mathbb{Z}$, set

$$Z((s_1, c_1), \dots, (s_d, c_d)) = \text{Li}_{s_1, \dots, s_d}(i^{c_1}, \dots, i^{c_d}).$$

Its integral word is

$$Z(\mathbf{s}; \mathbf{c}) = (-1)^d \mathcal{I} \left(0^{s_1-1} i^{-c_1} 0^{s_2-1} i^{-(c_1+c_2)} \cdots 0^{s_d-1} i^{-(c_1+\cdots+c_d)} \right). \quad (4.4)$$

This fixes both the color convention and every depth-dependent sign.

Consider the involution

$$\phi(t) = \frac{1-t}{1+t}.$$

It preserves the puncture set $\{0, \infty, 1, -1, i, -i\}$ and reverses the real interval. Direct differentiation gives

$$\phi^* \omega_a = \omega_{\phi(a)} - \omega_{-1} \quad (a \neq -1), \quad \phi^* \omega_{-1} = -\omega_{-1},$$

where $\phi(0) = 1$, $\phi(1) = 0$, $\phi(i) = -i$, and $\phi(-i) = i$. Denote this linear map on letters by τ . Changing variables and reversing the order of the simplex proves

$$\mathcal{I}(a_1 \cdots a_w) = (-1)^w \mathcal{I}(\tau(a_w) \cdots \tau(a_1)). \quad (4.5)$$

No tangential-basepoint correction occurs here: an original last letter other than 0 transforms into first letters other than 1, and an original first letter other than 1 transforms into last letters other than 0. Every term of the expansion therefore remains convergent. The broader use of octahedral symmetries to supplement standard level-four relations goes back to Zhao [12, 13]; the elementary involution above is sufficient for the present certificate.

Only one instance of (4.5) is needed. The word of $\text{Li}_{1,4}(i, 1)$ is $(-i)000(-i)$, and hence

$$D := \text{Im} \left\{ \mathcal{I}(e_{-i} e_0^3 e_{-i}) + \mathcal{I}((e_i - e_{-1})(e_1 - e_{-1})^3(e_i - e_{-1})) \right\} = 0. \quad (4.6)$$

Here e_a means the letter at a , multiplication is concatenation, and powers are powers in the noncommutative word algebra. The transformed product expands into exactly 32 integral words, with coefficients ± 1 . The first integral is the single-color double; the second group is converted to colored polylogarithms using (4.4).

4.3 The standard relation rows

The certificate uses two schemas, interpreted over $\mathbb{Q}$ before imaginary parts are taken. First, for two admissible multiple-polylogarithm index/color lists u, v , let $\mathbf{H}(u, v)$ be the integral shuffle product and $\mathbf{S}(u, v)$ the series stuffle product. Then

$$R_{u,v} := \text{Im}(\mathbf{H}(u, v) - \mathbf{S}(u, v)) = 0. \quad (4.7)$$

The same row is used when $u = ((1, 0))$ and v is admissible, with the standard single-divergence regularization. This point must not be suppressed: $\text{Li}_1(1)$ itself is divergent. Its shuffle and stuffle regularizations are both denoted by a formal variable T . A single leading $(1, 0)$ gives a polynomial of degree at most one in T , and the regularization comparison is the identity in degrees zero and one. Explicitly, its comparison operator satisfies

$$\rho(e^{Tu}) = \exp\left(\sum_{n \geq 2} \frac{(-1)^n \zeta(n)}{n} u^n\right) e^{Tu},$$

so $\rho(1) = 1$ and $\rho(T) = T$ [13, Theorem 2.9]. Subtracting the two product expansions cancels their common divergent word and leaves the convergent identity (4.7). This is the single-divergence case of regularized double shuffle [14, 13]. The verifier rejects all other divergent input patterns and checks that every surviving word is admissible.

Second, for a list $z = ((s_1, c_1), \dots, (s_d, c_d))$ with $c_j \in \{0, 1\}$ and with all terms below convergent, ordinary distribution is

$$Z(\mathbf{s}; 2\mathbf{c}) - 2^{w-d} \sum_{\epsilon \in \{0, 2\}^d} Z(\mathbf{s}; \mathbf{c} + \epsilon) = 0, \quad w = s_1 + \dots + s_d. \quad (4.8)$$

Indeed summing signs forces every nested summation index to be even, so the factor is exactly 2^{w-d} . The certificate multiplies some such relations by an admissible value of complementary weight and expands that product by integral shuffle. These are *lifted convergent distribution* rows; no regularized distribution formula is assumed. At weight one this includes the elementary seed $\text{Li}_1(-1) = \text{Li}_1(i) + \text{Li}_1(-i)$.

4.4 The exact certificate identity

In the formal real vector space whose generators are imaginary parts of convergent colored values of weight five, define

$$\begin{aligned} F = & 1120 \text{Im Li}_{4,1}(i, -1) + 480g_{4,1} + 480g_{3,2} + 1440g_{2,3} - 1536 \text{Im Li}_5(i) \\ & + 135 \text{Im}(\text{Li}_2(i) \text{Li}_3(1)) - 2240 \text{Im}(\text{Li}_4(i) \text{Li}_1(-1)). \end{aligned} \quad (4.9)$$

The two products in this expression are expanded by integral shuffle. Complex conjugation is imposed only by $\text{Im } Z(\mathbf{s}; -\mathbf{c}) = -\text{Im } Z(\mathbf{s}; \mathbf{c})$.

Lemma 4.2 (Finite rational certificate). *The accompanying file `S4_certificate.json` gives 911 rational numbers q_r and 911 standard-relation descriptors R_r such that the following is an exact equality of formal vectors:*

$$F = -1120D + \sum_{r=1}^{911} q_r R_r. \quad (4.10)$$

The rows consist of 713 convergent double-shuffle rows, 181 rows with one regularized weight-one factor, and 17 lifted convergent distribution rows.

Proof. The supplied program `verify_s4.py` rebuilds every row from the indices and colors in its descriptor. Integral shuffles are generated by choosing interleaving positions, series stuffles by the three defining recursive cases, and distribution by sign enumeration. The program also reconstructs the single duality seed and the target independently of their stored expansions. All coefficients are

integers or instances of Python’s exact `Fraction` type. Summing the left side minus the right side of (4.10) yields the empty coefficient dictionary, so every formal-word coefficient is exactly zero. The verifier neither reruns a rank calculation nor uses numerical polylogarithms, integer-relation searches, cached matrices, or modular arithmetic. Thus the certificate is a finite rational proof of the stated formal equality. $\square$

Proof of Theorem 4.1. Every R_r vanishes by (4.7) or (4.8), and $D = 0$ by (4.6). The certificate therefore gives $F = 0$. Now

$$\operatorname{Im} \operatorname{Li}_5(i) = \beta(5) = \frac{5\pi^5}{1536}, \quad \operatorname{Li}_3(1) = \zeta(3), \quad \operatorname{Li}_1(-1) = -\log 2.$$

Substituting these evaluations into (4.9) gives

$$1120 \operatorname{Im} \operatorname{Li}_{4,1}(i, -1) + 480g_{4,1} + 480g_{3,2} + 1440g_{2,3} - 5\pi^5 + 135G\zeta(3) + 2240\beta(4)\log 2 = 0.$$

Division by 1120 proves (4.2), and (4.3) proves (4.1). $\square$

Independent numerical audit. An independent real-quadrature evaluation uses

$$S_4 = \frac{1}{6} \int_0^1 \frac{\log^3 x \log(1+x^2)}{1+x^2} dx, \quad g_{a,b} = \frac{1}{(a-1)!} \int_0^1 (-\log x)^{a-1} \operatorname{Im} \frac{i \operatorname{Li}_b(ix)}{1-ix} dx.$$

At 70 decimal working digits it gives

$$S_4 = -0.0104934562973350434533567510062017519635106188773197615131294,$$

with discrepancy about 4.6×10^{-71} between the two sides. This numerical agreement is an audit of conventions, not a step in the proof; the proof is the exact certificate above.

Scope and further work. The existing two relations among $g_{4,1}, g_{3,2}, g_{2,3}, g_{1,4}$ may be used to shorten (4.1); they are prior results and are not counted as new here. The present proof establishes an upper bound on a specific constant’s expression complexity. It says nothing about a lower bound, linear independence, or the impossibility of a classical-polylogarithm reduction. A useful next problem is to compress the 911 standard rows into a short conceptual derivation, and to apply the same convergent involution to the analogous odd-weight mixed sums $S_6, S_8, \dots$

5 Exponential cancellation at proportional harmonic depth

The algebraic results above organize special arguments. We now turn to an analytic question left open by the earlier alternating-harmonic report [3]: how rapidly does its normalized sum tend to zero below the first-term transition? The case of proportional outer exponent and harmonic depth has an explicit rate function. This is a continuation of that report’s results, not another proof of its already established reflection identity.

For integers $h, n \geq 0$, define

$$e_h(n) = [u^h] \prod_{j=1}^n (1 + u/j), \quad T_{p,h}(a) = \sum_{n \geq 0} \frac{(-1)^n e_h(n)}{(n+a)^p}, \quad a, p > 0.$$

Thus $e_0(n) = 1$, $e_1(n) = H_n$, and $e_2(n) = (H_n^2 - H_n^{(2)})/2$. Ordinary convergence holds for $p > 0$; absolute convergence holds for $p > 1$. The Gaussian normalization is $S_{p,h} = 2^{-p}T_{p,h}(1/2)$. The positive Mellin integral

$$T_{p,h}(a) = \frac{(-1)^h}{h! \Gamma(p)} \int_0^\infty \frac{x^{p-1} e^{-ax} \log^h(1 + e^{-x})}{1 + e^{-x}} dx \quad (5.1)$$

follows from

$$\sum_{n \geq 0} e_h(n) z^n = \frac{(-\log(1 - z))^h}{h!(1 - z)}$$

by inserting an Abel factor, using the gamma integral, and taking its limit. The exponential decay at infinity and the bound $\log(1 + e^{-x}) \leq \log 2$ near zero justify the integral limit. The convergence and Abel-limit argument are also proved in [3, Section 3]. In particular, $(-1)^h T_{p,h}(a) > 0$.

Put

$$R_{p,h}(a) = (-1)^h h! (h + a)^p T_{p,h}(a). \quad (5.2)$$

The earlier report proves $0 < R_{p,h}(a) < 1$ and identifies the threshold $p \sim h \log h$. When $p/h - \log h \rightarrow -\infty$, it proves that $R_{p,h}(a) \rightarrow 0$. The following theorem quantifies this statement when p/h remains in a fixed positive compact interval.

5.1 The unique saddle and the rate function

Write

$$L(x) = \log(1 + e^{-x}), \quad w(x) = \frac{1}{(1 + e^x)L(x)} = -\frac{d}{dx} \log L(x), \quad x > 0. \quad (5.3)$$

Lemma 5.1. *The function w is strictly increasing, with*

$$\frac{1}{2 \log 2} < w(x) < 1.$$

For every $\alpha > 0$ there is exactly one $x_\alpha > 0$ satisfying

$$x_\alpha w(x_\alpha) = \alpha. \quad (5.4)$$

Moreover $\alpha < x_\alpha < 2\alpha \log 2$.

Proof. Set $t = e^{-x} \in (0, 1)$. Then $w = t/((1 + t) \log(1 + t))$ and

$$\frac{d}{dt} \log w = \frac{\log(1 + t) - t}{t(1 + t) \log(1 + t)} < 0.$$

Consequently w increases with x . Its endpoint limits give the bounds. The function $xw(x)$ has derivative $w + xw' > 0$, tends to zero at zero, and tends to infinity at infinity. This proves existence, uniqueness, and the stated enclosure. $\square$

For $x = x_\alpha$, set

$$\Phi_\alpha(y) = \alpha \log y + \log L(y), \quad (5.5)$$

$$A_\alpha = \frac{\alpha}{x^2} + w'(x) > 0, \quad (5.6)$$

$$I(\alpha) = -\alpha \log(x/\alpha) - \alpha - \log L(x), \quad (5.7)$$

$$C(\alpha, a) = \frac{\sqrt{\alpha/A_\alpha}}{x(1 + e^{-x})} \exp(a(\alpha - x)). \quad (5.8)$$

All of these functions are real analytic for $\alpha > 0$.

Theorem 5.2 (Proportional-depth asymptotics). *Let a and α range over fixed compact subsets of $(0, \infty)$, and put $p = \alpha h$. As the integer $h \rightarrow \infty$,*

$$R_{\alpha h, h}(a) = C(\alpha, a)e^{-hI(\alpha)}(1 + O(h^{-1})), \quad (5.9)$$

uniformly on those compact sets. In fact

$$R_{\alpha h, h}(a) = C(\alpha, a)e^{-hI(\alpha)} \left(1 + \frac{D(\alpha, a)}{h} + O(h^{-2}) \right), \quad (5.10)$$

where, with $x = x_\alpha$, $A = A_\alpha$, and $g(y) = e^{-ay}/(y(1 + e^{-y}))$,

$$D(\alpha, a) = \frac{g''(x)}{2Ag(x)} + \frac{\Phi_\alpha'''(x)g'(x)}{2A^2g(x)} + \frac{\Phi_\alpha''''(x)}{8A^2} + \frac{5\Phi_\alpha'''(x)^2}{24A^3} - \frac{\alpha a^2}{2} - \frac{1}{12\alpha}. \quad (5.11)$$

Every fixed further order in h^{-1} exists and is effectively obtained from derivatives of L and g at x_α and the Stirling series. The remainder at every fixed order is uniform on the same compact sets.

Proof. Equation (5.1) gives

$$R_{\alpha h, h}(a) = \frac{(h+a)^{\alpha h}}{\Gamma(\alpha h)} \int_0^\infty e^{h\Phi_\alpha(y)} g(y) dy.$$

We have $\Phi'_\alpha(y) = \alpha/y - w(y)$ and $\Phi''_\alpha(y) = -\alpha/y^2 - w'(y) < 0$. The sole critical point is therefore a strict global maximum. Its location stays in a fixed compact subinterval of $(0, \infty)$ as α varies in the prescribed compact set, and its curvature is bounded away from zero.

Here are explicit tail controls for the application of Laplace's method [17, §2.3(iii)] [17, Section 2.3(iii)]. Near zero, $\exp(h\Phi_\alpha(y)) \leq y^{h\alpha}(\log 2)^h$, while $g(y) = O(y^{-1})$ uniformly in a . Choosing the endpoint interval small enough makes its integral exponentially smaller than the maximum, uniformly in α . At infinity, $L(y) < e^{-y}$ implies $\Phi_\alpha(y) < \alpha \log y - y$; this supplies an integrable exponential bound uniform in the same compact parameters. On the remaining compact set outside a fixed neighborhood of x_α , strict concavity and compactness give a uniform negative gap from the maximum. These observations justify discarding both tails with an exponentially small relative error.

In the neighborhood, substitute $y = x + v/\sqrt{h}$. Taylor expansion, followed by Gaussian integration, gives

$$\begin{aligned} \int_0^\infty e^{h\Phi_\alpha(y)} g(y) dy &= e^{h\Phi_\alpha(x)} g(x) \sqrt{\frac{2\pi}{hA}} \left(1 + \frac{E}{h} + O(h^{-2}) \right), \\ E &= \frac{g''}{2Ag} + \frac{\Phi_\alpha'''g'}{2A^2g} + \frac{\Phi_\alpha''''}{8A^2} + \frac{5(\Phi_\alpha''')^2}{24A^3}, \end{aligned}$$

with all derivatives evaluated at x . Odd Gaussian moments vanish; the even moments $1/A$, $3/A^2$, and $15/A^3$ produce precisely these four terms. The analytic functions have uniformly bounded derivatives on a common neighborhood of the saddle. Taylor remainders there, together with the tail bounds just given, prove the claimed uniform error and also permit expansion to any prescribed order.

Finally Stirling's formula, uniformly for α in its compact set, and $\alpha h \log(1 + a/h) = \alpha a - \alpha a^2/(2h) + O(h^{-2})$ give

$$\frac{(h+a)^{\alpha h}}{\Gamma(\alpha h)} = \sqrt{\frac{\alpha h}{2\pi}} e^{h\alpha(1 - \log \alpha) + \alpha a} \left(1 - \frac{\alpha a^2/2 + 1/(12\alpha)}{h} + O(h^{-2}) \right).$$

Multiplying the two expansions proves (5.9)–(5.11). □

Remark 5.3 (Integer exponents and rounding). For integer p , set $\alpha = p/h$ in the theorem. If instead one fixes α_0 and rounds $\alpha_0 h$ to p , the $O(1/h)$ change of α must remain inside $e^{-hI(\alpha)}$: replacing it by $e^{-hI(\alpha_0)}$ changes the leading constant by a generally nontrivial bounded factor. This is a practical distinction in numerical comparisons.

5.2 Monotonicity, convexity, and matching limits

Proposition 5.4. *The rate I is positive, strictly decreasing, and strictly convex on $(0, \infty)$. More explicitly,*

$$I'(\alpha) = \log \frac{\alpha}{x_\alpha} = \log w(x_\alpha) < 0, \quad I''(\alpha) = \frac{w'(x_\alpha)}{w(x_\alpha)(w(x_\alpha) + x_\alpha w'(x_\alpha))} > 0. \quad (5.12)$$

Its endpoint limits are

$$I(0+) = -\log \log 2, \quad I(+\infty) = 0.$$

As $\alpha \rightarrow \infty$,

$$x_\alpha = \alpha + \frac{\alpha}{2}e^{-\alpha} + O(\alpha^2 e^{-2\alpha}), \quad (5.13)$$

$$I(\alpha) = \frac{1}{2}e^{-\alpha} - \frac{3\alpha + 5}{24}e^{-2\alpha} + O(\alpha^2 e^{-3\alpha}), \quad (5.14)$$

$$C(\alpha, a) = 1 - \left(\frac{(a + 1/2)\alpha}{2} + 1 \right) e^{-\alpha} + O_a(\alpha^2 e^{-2\alpha}). \quad (5.15)$$

Also $C(\alpha, a) \rightarrow 1/2$ as $\alpha \downarrow 0$.

Proof. Differentiating the optimized value in (5.7) cancels its x'_α terms by (5.4). This gives the first identity in (5.12). Differentiating $\alpha = xw(x)$ gives the second. Positivity also follows directly: $\log L(y) < -y$, so

$$\max_{y>0} \{\alpha \log(y/\alpha) + \alpha + \log L(y)\} < \max_{y>0} \{\alpha \log(y/\alpha) + \alpha - y\} = 0.$$

The strict inequality is valid because the first maximum is attained; at its maximizing point the second expression is at most zero.

The small- α limits follow from $x_\alpha/\alpha \rightarrow 2 \log 2$ and $A_\alpha \sim \alpha/x_\alpha^2$. At infinity, put $t = e^{-y}$ and use

$$\log L(y) = -y - \frac{t}{2} + \frac{5t^2}{24} + O(t^3), \quad w(y) = 1 - \frac{t}{2} + \frac{5t^2}{12} + O(t^3).$$

Substitution in (5.4) first gives (5.13). Taylor expansion of (5.7) around $y = \alpha$, retaining the quadratic displacement, gives (5.14). Expansion of (5.8) and $A_\alpha = \alpha/x_\alpha^2 + w'(x_\alpha)$ gives (5.15). The remainders stated above follow from analyticity of $\log(\log(1+t)/t)$ for t near zero, with $x_\alpha - \alpha = O(\alpha e^{-\alpha})$. $\square$

The rate formula explains the scale of cancellation even when the terms of the original alternating sum are much larger than its value. For example, $I(1) = 0.15091647898554\dots$ and $C(1, 1/2) = 0.66325765726435\dots$. Thus

$$R_{h,h}(1/2) \sim 0.66325765726435\dots e^{-0.15091647898554\dots h}.$$

This is a proved exponential rate, rather than just a limit of zero.

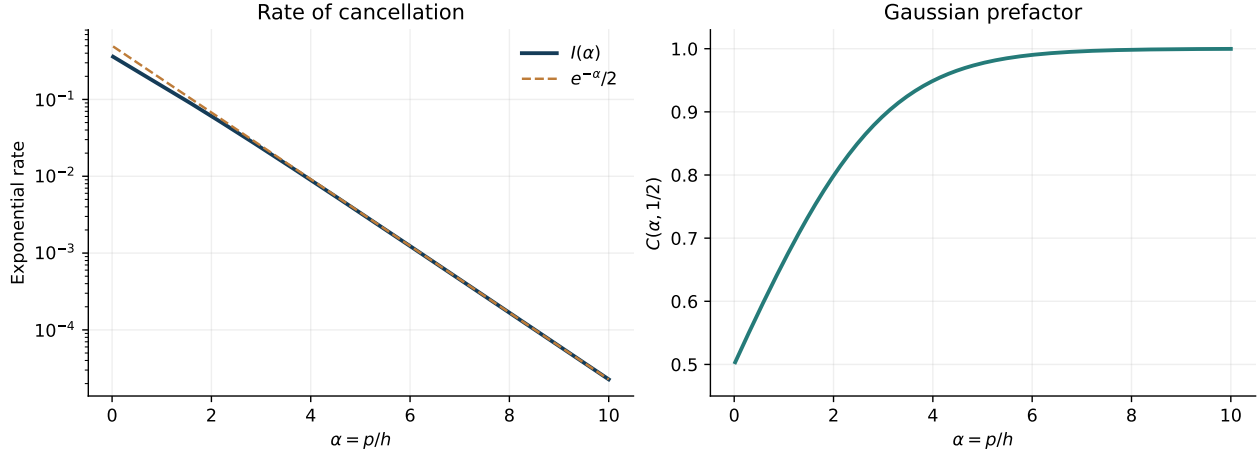


Figure 1: The proved rate $I(\alpha)$ and prefactor $C(\alpha, 1/2)$ from Section 5. The dashed curve is the leading large- α approximation $e^{-\alpha}/2$. This plot illustrates the exact formulas; it supplies no additional proof or error bound.

The large- α expansions also explain why the transition parameter in the previous report was $\lambda = he^{-\alpha}/2$. Formally putting $\alpha = \log h + c$ in $hI(\alpha)$ gives the leading value $e^{-c}/2$. This agrees with the independently proved transition $R \rightarrow \exp(-e^{-c}/2)$ in [3]. Such matching is a consistency calculation: the compact- α theorem proved here does *not* by itself establish a uniform error for growing α . Obtaining a single effective formula across that expanding range remains a precise further problem.

5.3 A probability law at the saddle

There is a useful formulation independent of the alternating series. Normalize the positive integrand in (5.1), with $p = \alpha h$, to a probability density and call the resulting random variable X_h . Then

$$\sqrt{hA_\alpha}(X_h - x_\alpha) \implies N(0, 1). \quad (5.16)$$

Indeed, the same change of variables used in the proof of Theorem 5.2, its uniform Gaussian domination on a shrinking neighborhood, and its exponentially negligible tails prove convergence against every bounded continuous test function. The associated rate for X_h is $\Phi_\alpha(x_\alpha) - \Phi_\alpha(y)$. This identifies which part of the positive integral carries the very small alternating sum and gives a stable center for numerical quadrature.

6 Reflected log-gamma moments and their late coefficients

Set $f(x) = \log \Gamma(x)$ for $0 < x < 1$, and define

$$M_{n,m} = \int_0^1 f(x)^n f(1-x)^m dx, \quad n, m \in \mathbb{Z}_{\geq 0}.$$

These integrals already occur as $T_{n+m,n}$ in Amdeberhan et al. [16, Section 5]. Their symmetry $M_{n,m} = M_{m,n}$ and the identities obtained by expanding Euler's reflection formula are therefore classical. The asymptotic question here is different: keep the reflected exponent m fixed and let

n tend to infinity. The factor $f(1-x)^m$ vanishes to order m at the endpoint that produces the factorial growth. This shifts the first exponential scale from 1 to $m+1$ without removing the eventual divergence of the full expansion. Theorem 6.2 identifies its late terms, including their signs and the first relative correction.

6.1 The meromorphic transform and all finite asymptotic orders

Put

$$B(x) = \log \Gamma(1-x) = \gamma x + \sum_{j \geq 2} \frac{\zeta(j)}{j} x^j, \quad |x| < 1,$$

where γ is Euler's constant. For $k \geq 1$ define

$$A_{k,m} = \frac{1}{k} [x^{k-1}] \exp\{kB(-x)\} B(x)^m. \quad (6.1)$$

The convention $B(x)^0 = 1$ is in force. In particular $A_{k,m} = 0$ when $k \leq m$. Every coefficient is a finite polynomial over $\mathbb{Q}$ in $\gamma, \zeta(2), \dots, \zeta(k-1)$.

Theorem 6.1 (Reflected moment expansion). *For each fixed integer $m \geq 0$, the function*

$$F_m(z) = \int_0^1 \Gamma(x)^z f(1-x)^m dx$$

is holomorphic for $\operatorname{Re} z < m+1$, has Taylor expansion

$$F_m(z) = \sum_{n \geq 0} \frac{M_{n,m}}{n!} z^n \quad (|z| < m+1),$$

and extends meromorphically to $\mathbb{C}$. Its only possible poles are simple poles at integers $k \geq m+1$, with

$$\operatorname{Res}_{z=k} F_m(z) = -k A_{k,m}. \quad (6.2)$$

The first pole is present. For every fixed integer $K \geq m+1$,

$$\frac{M_{n,m}}{n!} = \sum_{k=m+1}^K \frac{A_{k,m}}{k^n} + O_{m,K}((K+1)^{-n}), \quad n \longrightarrow \infty. \quad (6.3)$$

Consequently

$$M_{n,m} \sim \frac{\gamma^m n!}{(m+1)^{n+1}}. \quad (6.4)$$

Proof. As $x \downarrow 0$, $f(x) = -\log x + O(x)$ and $B(x) = \gamma x + O(x^2)$. At the other endpoint, $f(x) = O(1-x)$ and $f(1-x) = O(1+|\log(1-x)|)$. These estimates prove absolute convergence and compact domination in the asserted half-plane, including after any number of differentiations in z . They also justify the Taylor expansion near zero.

Fix $0 < a < 1$ and an integer $J \geq 1$. Taylor expansion in x gives

$$\Gamma(1+x)^z B(x)^m = \sum_{j=0}^{J-1} b_{j,m}(z) x^j + x^J E_{J,m}(x, z),$$

where each $b_{j,m}$ is an entire function of z , and the remainder is uniformly bounded for $0 \leq x \leq a$ and z in compact sets. The integral over $[0, a]$ is therefore

$$\sum_{j=0}^{J-1} \frac{b_{j,m}(z)a^{j+1-z}}{j+1-z} + \int_0^a x^{J-z} E_{J,m}(x, z) dx.$$

The last integral is holomorphic for $\operatorname{Re} z < J + 1$. The integral over $[a, 1]$ is entire, since its logarithmic singularity at 1 is integrable uniformly for z in compact sets. Letting J increase gives the continuation. At $z = k$ its residue is $-b_{k-1,m}(k) = -kA_{k,m}$. Since $B(x)^m$ starts with $\gamma^m x^m$, the first nonzero coefficient is $A_{m+1,m} = \gamma^m/(m+1) > 0$.

For the asymptotic assertion, subtract the polar functions

$$\sum_{k=m+1}^{K+1} \frac{kA_{k,m}}{k-z}$$

from $F_m(z)$. The difference is holomorphic on $|z| < K + 2$. Cauchy's estimate on any circle $K + 1 < R < K + 2$ bounds its n th Taylor coefficient by $O_{m,K,R}(R^{-n})$. Restoring the term at $K + 1$ proves (6.3), even if that particular coefficient happens to vanish. Taking $K = m + 1$ proves (6.4). $\square$

The first three coefficients are explicit. Write $k = m + 3$ in the last line:

$$\begin{aligned} A_{m+1,m} &= \frac{\gamma^m}{m+1}, \\ A_{m+2,m} &= \frac{\gamma^{m-1}}{m+2} \left(\frac{m\zeta(2)}{2} - (m+2)\gamma^2 \right), \\ A_{m+3,m} &= \frac{\gamma^m}{k} \left\{ \frac{m\zeta(3)}{3\gamma} + \frac{m(m-1)\zeta(2)^2}{8\gamma^2} + \frac{k(1-m)\zeta(2)}{2} + \frac{k^2\gamma^2}{2} \right\}. \end{aligned} \tag{6.5}$$

These displays also apply to $m = 0$, since $\gamma > 0$ and factors of m remove the superfluous terms. For example,

$$\begin{aligned} \frac{M_{n,1}}{n!} &= \frac{\gamma}{2^{n+1}} + \frac{\zeta(2)/2 - 3\gamma^2}{3^{n+1}} + \frac{\zeta(3)/3 + 8\gamma^3}{4^{n+1}} + O(5^{-n}), \\ \frac{M_{n,2}}{n!} &= \frac{\gamma^2}{3^{n+1}} + \frac{\gamma\zeta(2) - 4\gamma^3}{4^{n+1}} + O(5^{-n}). \end{aligned}$$

The second displayed correction is positive, whereas the corresponding correction for $m = 1$ is negative. Thus strict alternation from the first term cannot be imported from the unreflected family. The two signs can, for instance, be checked using the elementary enclosures $0.577 < \gamma < 0.578$ and $1.644 < \zeta(2) < 1.646$. Eventual alternation has a different, precise rule.

6.2 Late coefficients and a universal inverse-gamma scale

Let $\phi(u) = \Gamma(1-u)$ and $D = u d/du$. Define $\tau \in (0, 1)$ by

$$D \log \phi(\tau) = -\tau \psi(1-\tau) = 1, \tag{6.6}$$

and put

$$\rho = \frac{\tau}{\phi(\tau)}, \quad v = D^2 \log \phi(\tau) = 1 + \tau^2 \psi'(1-\tau), \quad C = \frac{\tau}{\sqrt{2\pi v}}, \quad b = -\log \Gamma(1+\tau). \tag{6.7}$$

The equation for τ has a unique solution: its left side equals $\gamma u + \sum_{j \geq 2} \zeta(j) u^j$, which is strictly increasing from zero to infinity. Also $0 < \rho < 1$ and $b > 0$, the latter by strict log-convexity of Γ between 1 and 2. Numerically, $\tau = 0.5040830082\dots$ and $\rho = 0.2821158090\dots$. Only the exact definitions are used below.

Theorem 6.2 (Late reflected coefficients). *For every fixed nonnegative integer m ,*

$$A_{k,m} = (-1)^{k+m-1} C b^m \rho^{-k} k^{-3/2} \left(1 + \frac{\beta_m}{k} + O_m(k^{-2}) \right). \quad (6.8)$$

There is a full asymptotic expansion in integer powers of $1/k$. To specify the first correction, set $G(u) = \log \Gamma(1+u)$, $H_m(u) = G(u)^m$, and $\kappa_j = D^j \log \phi(\tau)$. All quantities on the right below are evaluated at $u = \tau$:

$$\beta_m = -\frac{(D+1)^2 H_m}{2v H_m} + \frac{\kappa_3 (D+1) H_m}{2v^2 H_m} + \frac{\kappa_4}{8v^2} - \frac{5\kappa_3^2}{24v^3}. \quad (6.9)$$

In particular, every sufficiently large possible pole in Theorem 6.1 is present. More explicitly,

$$\text{Res}_{z=k} F_m(z) = (-1)^{k+m} C b^m \rho^{-k} k^{-1/2} \left(1 + \frac{\beta_m}{k} + O_m(k^{-2}) \right),$$

so the eventual residue sign is $(-1)^{k+m}$. For every fixed real n , the formal series $\sum_{k \geq m+1} A_{k,m}/k^n$ diverges because its terms do not tend to zero.

Proof. Replacing x by $-u$ in (6.1) gives

$$A_{k,m} = \frac{(-1)^{k-1}}{k} [u^{k-1}] \phi(u)^k H_m(u).$$

The function H_m is holomorphic for $|u| < 1$. If $\phi(u) = \sum_{j \geq 0} a_j u^j$, all the a_j are strictly positive, since $\log \phi$ has positive coefficients. Thus

$$p_j = \frac{a_j \tau^j}{\phi(\tau)}$$

is a probability distribution with exponential moments, mean one, variance $v > 0$, and positive mass at both 0 and 1. It follows that the function

$$Q(t) = \frac{\phi(\tau e^{it})}{\phi(\tau)} e^{-it}$$

has modulus strictly less than one for $0 < |t| \leq \pi$, while

$$\log Q(t) = -\frac{v}{2} t^2 - \frac{i\kappa_3}{6} t^3 + \frac{\kappa_4}{24} t^4 + O(t^5)$$

near zero. Cauchy's coefficient formula becomes

$$A_{k,m} = \frac{(-1)^{k-1} \tau \rho^{-k}}{2\pi k} \int_{-\pi}^{\pi} Q(t)^k e^{it} H_m(\tau e^{it}) dt. \quad (6.10)$$

Fix a sufficiently small $\delta > 0$. On $\delta \leq |t| \leq \pi$ the integral is $O(q^k)$ for some $q < 1$. On $k^{-2/5} \leq |t| \leq \delta$ it is $O(e^{-ck^{1/5}})$, because $|Q(t)| \leq e^{-ct^2}$. The analytic amplitude is bounded on the

whole circle. On the remaining interval set $t = y/\sqrt{k}$. Gaussian domination and Taylor expansion to any fixed order, with an analytic remainder, give a full expansion. Odd terms integrate to zero, so only integer powers of $1/k$ occur. The leading integral is

$$H_m(\tau)\sqrt{\frac{2\pi}{vk}} = (-1)^m b^m \sqrt{\frac{2\pi}{vk}},$$

which proves the leading term in (6.8).

For the first correction, the amplitude derivatives at $t = 0$ are H_m , $i(D+1)H_m$, and $-(D+1)^2 H_m$. The Gaussian moments of degrees 2, 4, and 6 contribute, respectively, the amplitude's quadratic term, the amplitude-cubic cross term together with the quartic phase term, and the square of the cubic phase term. Their normalized sum is exactly (6.9). Expanding two further even orders gives a relative $O_m(k^{-2})$ remainder after the displayed correction. Since $Cb^m > 0$ and $\rho < 1$, the assertions about nonvanishing, signs, and divergence follow at once. $\square$

This proof uses the circle $|u| = \tau$ and a strict modulus inequality; it does not presume a classification of all complex critical values of the reciprocal gamma function. The same circle governs every fixed reflected exponent. With $\alpha = -\log \rho$, the continuous minimum of the leading term magnitude $\rho^{-k} k^{-n-3/2}$ occurs at $k = (n + 3/2)/\alpha$. A least-term calculation alone does not bound the remainder. The next result supplies an explicit bound with a growing number of terms.

6.3 An effective remainder despite the initial sign changes

Theorem 6.3 (Explicit reflected truncation bound). *Let $m \geq 0$, $n \geq 1$, and $K \geq 1$ be integers, put $\alpha = -\log \rho$ and $T_m = \tau B(\tau)^m$, and suppose $K\alpha < n$. With*

$$R_{n,K}^{(m)} = \frac{M_{n,m}}{n!} - \sum_{k=1}^K \frac{A_{k,m}}{k^n}, \quad \Gamma(n, s) = \int_s^\infty t^{n-1} e^{-t} dt,$$

one has the entirely explicit estimate

$$\begin{aligned} |R_{n,K}^{(m)}| &\leq \frac{\alpha^n}{n!} \left(m! + \frac{nT_m}{n - K\alpha} \right) \\ &\quad + \frac{T_m \rho^{-K-1}}{(K+1)^n} \frac{\Gamma(n, (K+1)\alpha)}{\Gamma(n)}. \end{aligned} \quad (6.11)$$

If $n \geq \max\{2, 4\alpha^2\}$ and $K = \lfloor n/\alpha - \sqrt{n} \rfloor$, then $K \geq 1$ and

$$|R_{n,K}^{(m)}| \leq \left(\frac{m!}{\sqrt{2\pi n}} + \frac{T_m}{\alpha\sqrt{2\pi}} + T_m e^{2\alpha^2} \right) \left(\frac{e\alpha}{n} \right)^n. \quad (6.12)$$

This bound holds for every m , but no useful uniform relative error is asserted when m grows with n .

Proof. The function f decreases from infinity to zero on $(0, 1]$: $\psi'(x) > 0$ and $\psi(1) = -\gamma < 0$. Let $x(y)$ denote its inverse, and set $W_m(x) = \int_0^x B(t)^m dt$. Tonelli's theorem applied to $f(x)^n = n \int_0^{f(x)} y^{n-1} dy$ gives

$$\frac{M_{n,m}}{n!} = \frac{1}{\Gamma(n)} \int_0^\infty y^{n-1} W_m(x(y)) dy. \quad (6.13)$$

Let $h(q)$ be the local solution of $h = q\phi(h)$ with $h(0) = 0$. Lagrange inversion gives positive coefficients

$$h(q) = \sum_{k \geq 1} c_k q^k, \quad c_k = \frac{1}{k} [u^{k-1}] \phi(u)^k.$$

The $m = 0$ case of Theorem 6.2 proves that its radius is ρ and that the series converges absolutely on $|q| \leq \rho$. The map $u \mapsto u/\phi(u)$ increases strictly from 0 to ρ on $[0, \tau]$, so inversion along this interval and continuity imply $h(\rho) = \tau$. No other inverse branch is used.

Lagrange–Bürmann inversion now gives

$$W_m(x(y)) = \sum_{k \geq 1} A_{k,m} e^{-ky} \quad (y \geq \alpha).$$

Indeed $x(y) = -h(-e^{-y})$ initially for large y , and then throughout $y > \alpha$ by analytic continuation along the strictly monotone real inverse. The endpoint follows by absolute convergence. To prove that convergence and obtain its quantitative majorant, define

$$d_{k,m} = \frac{1}{k} [u^{k-1}] \phi(u)^k B(u)^m \geq 0.$$

Coefficientwise absolute values in $B(-u)^m$ are $B(u)^m$. Consequently $|A_{k,m}| \leq d_{k,m}$, while another application of Lagrange–Bürmann gives

$$\sum_{k \geq 1} d_{k,m} \rho^k = W_m(h(\rho)) = W_m(\tau) \leq \tau B(\tau)^m = T_m.$$

Here B has positive Taylor coefficients and is increasing on $[0, \tau]$. Thus on $y \geq \alpha$,

$$\left| W_m(x(y)) - \sum_{k=1}^K A_{k,m} e^{-ky} \right| \leq T_m \rho^{-K-1} e^{-(K+1)y}.$$

Its integral in (6.13) is the second term of (6.11).

On $0 \leq y \leq \alpha$, use $0 \leq W_m(x(y)) \leq M_{m,0} \leq m!$. The last inequality follows from $0 \leq \log \Gamma(x) \leq -\log x$, since log-convexity gives $\Gamma(1+x) \leq 1$ for $0 \leq x \leq 1$. Also, for $k \leq K$,

$$\int_0^\alpha y^{n-1} e^{-ky} dy \leq \frac{\alpha^n e^{-k\alpha}}{n - k\alpha}.$$

This follows by differentiating $y^n e^{-ky}$ and using $n - ky \geq n - k\alpha > 0$. Summing with $\sum |A_{k,m}| \rho^k \leq T_m$ proves the first term of (6.11).

For the stated choice of K , $n - K\alpha \geq \alpha\sqrt{n}$ and $\theta = (K+1)\alpha/n$ lies in $[1/2, 1]$. The inequality $\theta - 1 - \log \theta \leq 2(1 - \theta)^2$ gives

$$\rho^{-K-1} (K+1)^{-n} \leq e^{2\alpha^2} (e\alpha/n)^n.$$

Finally, $\Gamma(n, s) \leq \Gamma(n)$ and $n! \geq \sqrt{2\pi n} (n/e)^n$ yield (6.12). □

6.4 An exact identity coupling all reflected residues

The coefficients across different m satisfy a finite identity in which Euler's constant and all odd zeta values cancel.

Proposition 6.4 (Reflection sum for the residues). *For every integer $k \geq 1$,*

$$\sum_{m=0}^{k-1} \frac{k^m}{m!} A_{k,m} = \begin{cases} 0, & k \text{ even}, \\ \frac{\pi^{k-1}}{k 2^{k-1}} \binom{k-1}{(k-1)/2}, & k \text{ odd}. \end{cases} \quad (6.14)$$

Proof. Because $B(x)^m$ has order m , extending the sum over m to infinity does not affect $[x^{k-1}]$. Euler's reflection formula gives

$$\sum_{m=0}^{k-1} \frac{k^m}{m!} A_{k,m} = \frac{1}{k} [x^{k-1}] (\Gamma(1+x)\Gamma(1-x))^k = \frac{1}{k} [x^{k-1}] \left(\frac{\pi x}{\sin \pi x} \right)^k.$$

The coefficient is zero for even k . For odd k , compute it as the residue at $x = 0$ of $(\pi/\sin \pi x)^k dx$. The local change of variable $t = \sin \pi x$ gives

$$[x^{k-1}] \left(\frac{\pi x}{\sin \pi x} \right)^k = \pi^{k-1} [t^{k-1}] (1-t^2)^{-1/2},$$

and the binomial expansion proves (6.14). $\square$

For example, the cases $k = 2, 3$ read $A_{2,0} + 2A_{2,1} = 0$ and $A_{3,0} + 3A_{3,1} + \frac{9}{2}A_{3,2} = \pi^2/6$. They are exact identities among the asymptotic coefficients, rather than convergent sums of the moment expansions. No summation over the late index k is used.

6.5 A recorded log-sine conjecture has a formal proof

There is also a small but useful correction to the logical status of a claim in the literature. The author preprint of [16] labels a coefficient identity as Conjecture 5.5 and motivates it using transcendence assumptions on $\log 2$ and $\log \pi$. For the canonical formal polynomials supplied by the beta integral, the coefficient identity needs none of those assumptions. Under the preprint's assumptions these formal coefficients coincide with its uniquely specified numerical coefficients, so the result also proves that stated implication. We do not assert uniqueness of arbitrary numerical expressions in zeta values and logarithms, nor priority over any later treatment.

Define polynomials $P_N(X)$ by

$$\sum_{N \geq 0} P_N(X) \frac{z^N}{N!} = \exp \left\{ Xz + \sum_{j \geq 2} \frac{(1-2^{1-j})\zeta(j)}{j} z^j \right\}. \quad (6.15)$$

Proposition 6.5 (Unconditional coefficient and translation identities). *For $N \geq 0$,*

$$\int_0^1 [-\log \sin(\pi x)]^N dx = P_N(\log 2), \quad (6.16)$$

$$[X^j] P_N(X) = \binom{N}{j} P_{N-j}(0), \quad 0 \leq j \leq N,$$

$$\sum_{j=0}^N \binom{N}{j} M_{j,N-j} = P_N(\log(2\pi)).$$

Moreover, the coefficients admit the finite partition formula

$$P_N(X) = N! \sum_{r_1+2r_2+\dots+N r_N=N} \frac{X^{r_1}}{r_1!} \prod_{j=2}^N \frac{1}{r_j!} \left(\frac{(1-2^{1-j})\zeta(j)}{j} \right)^{r_j}. \quad (6.17)$$

Proof. For $\operatorname{Re} z < 1$, the beta integral gives

$$\int_0^1 \sin(\pi x)^{-z} dx = \frac{\Gamma((1-z)/2)}{\sqrt{\pi} \Gamma(1-z/2)}.$$

Expanding its logarithm at zero gives the right side of (6.15) with $X = \log 2$; its degree- j coefficient for $j \geq 2$ is $(1 - 2^{1-j})\zeta(j)/j$. Extracting Taylor coefficients proves (6.16). The factor e^{Xz} gives $[X^j]P_N = \binom{N}{j}P_{N-j}(0)$ by formal coefficient extraction. Euler's reflection formula turns the sum of mixed moments into the integral of $(\log \pi - \log \sin \pi x)^N$, which translates X by $\log \pi$. Expanding each exponential factor separately proves (6.17). All formal operations in a fixed degree involve only finitely many coefficients. $\square$

The constant-column generating function has the additional factorization

$$\sum_{N \geq 0} P_N(0) \frac{z^N}{N!} = \sqrt{\frac{\tan(\pi z/2)}{\pi z/2}} \exp \left\{ \sum_{\substack{j \geq 3 \\ j \text{ odd}}} \frac{(1 - 2^{1-j})\zeta(j)}{j} z^j \right\}, \quad (6.18)$$

where the square root is the formal branch with constant term one. Indeed, multiplying the left side by its value at $-z$ and applying the reflection formula yields $\tan(\pi z/2)/(\pi z/2)$. The even logarithmic part is therefore the displayed square root. This makes the multiplicative factorization of its formal odd-zeta monomial coefficients explicit, while the even-zeta contribution has an elementary trigonometric generating function.

Verification and remaining questions. The accompanying script code/verify_reflected_moments.py checks the three coefficient formulas by exact symbolic arithmetic, checks finite reflection sums and the formal Appell formulas, and compares independent quadrature with the fixed-order expansion. It also compares the late coefficients with (6.8) and (6.9). The numerical calculations are diagnostics; the preceding analytic and formal arguments are the proofs. Two substantive next questions are uniformity when m increases with n and a signed remainder at the least term for each reflected exponent. In the former regime the endpoint approximation $B(x)^m \approx \gamma^m x^m$ need not be uniform, so the fixed- m theorem does not settle the simultaneous large-exponent problem.

7 Manuscript audit and proposed corrections

The audit is tied to the fixed revision in Section 1. Some objections recorded in earlier research packages have already been corrected in that revision. They should not be reported again as newly discovered defects. The following changes are supported by the proofs and direct source inspection in this article.

7.1 A confirmed even-weight normalization error

In chapters/03-algebraic.tex, equation bloch:eq:Pm defines

$$P_m(x) = \operatorname{Re}_m \left(\sum_{j=0}^{m-1} \frac{2^j B_j}{j!} (\log |x|)^j \operatorname{Li}_{m-j}(x) \right), \quad \operatorname{Re}_m = \begin{cases} \operatorname{Re}, & m \text{ odd}, \\ \operatorname{Im}, & m \text{ even}. \end{cases}$$

For $0 < x < 1$, every term inside the parentheses is real. Consequently

$$P_{2j}(x) = 0 \quad (0 < x < 1). \quad (7.1)$$

This is also an immediate consequence of the conjugation law printed in the same chapter. The paragraph immediately following `bloch:eq:goldenP3` and `bloch:eq:sgP3` nevertheless says that the higher golden ladders become the stated zeta multiples under the same modification, and explicitly includes the weight-six multiple. That assertion is incompatible with (7.1).

The correction is local. Keep the two displayed weight-three formulas. Explain that their lower-weight logarithmic terms cancel under P_3 . At other odd weights, carry out the full modified-polylogarithm calculation and state the proof status of the resulting formula. At even weight, keep the ordinary Li_{2j} identity in its ordinary normalization; it cannot become a nonzero real zeta multiple of this P_{2j} at real arguments. A different real-valued normalization would need to be explicitly defined and checked. The file `integration/proposed_P_even_correction.tex` contains replacement prose ready to paste into the indicated location. The associated patch is restricted to this confirmed error.

This correction does not disprove the numerical ordinary-polylogarithm formulas at weights six and eight. It corrects their asserted interpretation. The distinction matters because a vanishing single-valued evaluation cannot retain the nonzero zeta value that the paragraph assigns to it.

7.2 Three numerical statements can now be promoted

The equation `gauss:eq:S4-closed` in `chapters/04-depth.tex` is an exact theorem by Theorem 4.1. The surrounding paragraph currently places the whole displayed group in a numerically supported category unless an analytic derivation is supplied. Add a specific reference to the present proof and certificate for this equation. This does not automatically promote neighboring identities whose proofs are not given here.

In the subsection on the reciprocal minimal Pisot number in `chapters/03-algebraic.tex`, the displayed plastic dilogarithm and trilogarithm formulas now have the explicit proofs supplied in the preceding sections. Their normalization agrees exactly with the manuscript: $6/4 = 3/2$, $3/9 = 1/3$, and $6\lambda^3/3! = \lambda^3$. The adjacent plastic tetralogarithm, which introduces the exponents 7 and 14, is a separate identity. No proof of that formula follows merely by citing the two lower-weight results.

Several higher golden formulas are described using language such as “hold” or “correct” immediately after residual calculations. Where no independently traceable analytic proof is supplied, prefer “numerically verified to the stated precision.” A residual, even an extremely small one reproduced by different evaluators, establishes neither an exact equality nor its uniqueness. This is a proof-status clarification, not an assertion that those formulas are false.

7.3 What the classification does and does not imply

The pure-complement classification is exhaustive. It permits the exact statement that the reciprocal plastic family is the only real positive cyclic-power family with two different complement pairs. It does not bound the maximal weight of a polylogarithm ladder. A relation such as

$$\prod_j (1 - r^j)^{n_j} = r^N, \quad n_j, N \in \mathbb{Z},$$

may use many factors that are not themselves pure powers of r . Nor does the classification exclude additional units or nonmonomial arguments in the field. These possibilities are central to higher ladders and must remain distinct from the theorem proved here.

The degree bound is also specific: it makes the search for $r^a + r^b = 1$ complete at fixed degree. A finite coefficient-height box for arbitrary algebraic arguments has no comparable completeness consequence without an additional height theorem.

7.4 Analytic qualifications needed for integration

There are three limits on the gamma and harmonic statements worth preserving verbatim when they are integrated.

First, the late reflected-coefficient series is divergent for every fixed n . Its finite truncations form an asymptotic expansion, and the growing-order theorem supplies a rigorously controlled truncation. The displayed bound is unsigned; eventual alternation of coefficients does not establish the sign of the optimized remainder.

Second, m is fixed in the reflected-moment theorems. Reflection symmetry $M_{n,m} = M_{m,n}$ does not turn these statements into a theorem when both indices grow proportionally. In the harmonic theorem, $\alpha = p/h$ stays in a compact subset of $(0, \infty)$. Substitution of $\alpha = \log h + c$ into its error term is not justified by that compact-uniform assertion alone.

Third, the Appell identity concerns canonical formal coefficients defined by the generating function. It gives an unconditional formal version of the inspected author preprint's Conjecture 5.5. It does not prove transcendence or uniqueness of arbitrary numerical expressions involving logarithms and zeta values. The attribution is to the explicitly inspected 2010 preprint, without asserting that the formulation remained open in every later version of the literature.

7.5 Suggested integration sequence

The new material can be integrated in four independently reviewable units. Put the complementary-power classification and the two plastic proofs beside the existing algebraic ladders. Put the S_4 theorem, its word convention, and its certificate beside the Gaussian weight-five identities. Add the proportional-depth theorem to the alternating-harmonic report and cross-reference it from the depth chapter. Add the reflected moment results to the gamma-moment material, retaining the earlier unreflected theorem as the special case $m = 0$.

The article and all certificates are self-contained within this package. The supplied narrow patch uses the audited source context and can be reviewed before application. No repository branch, commit, or remote publication was created as part of this delivery.

8 Further research questions

The completed results suggest specific next steps. We distinguish a request for a sharper proof from a conjecture about the mathematics; the absence of a reduction in a finite search is never used as an independence conjecture by itself.

8.1 Gaussian identities and proof compression

1. Compress the S_4 certificate. Find a short derivation of Theorem 4.1 from a small collection of named functional identities. The present certificate is a complete proof, but its 911 standard rows obscure which intermediate values are essential. A useful computational formulation is sparse exact elimination with penalties on depth, color changes, and coefficient height. Every proposed

compression should be replayed by the independent verifier. This is a concrete optimization problem with an already certified target.

2. Determine the higher even-index mixed reductions. Apply the convergent involution to $\text{Li}_{1,2m}(i, 1)$ and to other low-depth seeds at weights $2m + 1$. For $m = 3, 4$, ask whether S_{2m} can be expressed using the same-point doubles $\text{Im Li}_{a,2m+1-a}(i, 1)$ and an explicitly specified algebra of lower-weight products. The phrase “explicitly specified” is essential: changing the allowed basis changes the problem. A positive answer requires a rational certificate; a negative answer in a chosen formal relation system only describes that system.

3. Organize the octahedral relations by representations. The puncture set $\{0, \infty, \pm 1, \pm i\}$ has a rich symmetry group. Its action on iterated-integral words suggests decomposing the formal weight space before elimination. Determine which symmetry component contains the S_4 target and why one additional convergent seed suffices for this particular calculation. The aim is a theorem about the relation module, with motivic and numerical period claims stated separately.

4. Formalize the finite certificate. The standard-library verifier has a deliberately small interface: word shuffles, stuffles, sign distribution, conjugation, and one Möbius substitution. These are suitable for translation into a proof assistant. The mathematical task is to connect the formal relation schemas to the analytic definitions, including the single-divergence regularization. Checking a JSON residual alone would formalize only the linear algebra, not that analytic connection.

8.2 Algebraic bases and higher ladders

5. Prove the plastic tetralogarithm. The adjacent manuscript formula, in reduced coefficients, is

$$0 = -71 \text{Li}_4(q) - 15 \text{Li}_4(q^2) + 39 \text{Li}_4(q^3) - 17 \text{Li}_4(q^5) + 16 \text{Li}_4(q^7) - \text{Li}_4(q^{14}) \\ + 51\zeta(4) + \frac{335}{4}\lambda^4 - 42\zeta(2)\lambda^2, \quad q^3 + q^2 = 1, \quad \lambda = \log q. \quad (8.1)$$

This is an inherited, numerically supported target, not a new conjecture introduced by this article. The low-weight proofs suggest searching for a specialization of a tetralogarithm functional equation whose arguments close after introducing q^7 and q^{14} . A symbol or cobracket calculation can identify candidates, but the remaining constant and branch terms still need proof.

6. Move from individual complements to product relations. Classify, or effectively enumerate at bounded support and exponent height, relations $\prod_j (1 - r^j)^{n_j} = r^N$ in a specified number field. The theorem in this article shows that individual pure complements have very limited redundancy. Higher ladders therefore invite a systematic study of product relations and of the additional units they introduce. Exact ideal valuations, units, and cyclotomic factorizations offer a natural algebraic starting point.

7. Refine the degree count. The exact formula for $N(D)$ reduces its error term beyond $D^2/2$ to an explicit sum involving $e(\lfloor D/d \rfloor + 1)$ and $e(\lfloor D/d \rfloor + 2)$. Determine its secondary asymptotic behavior, including any $D \log D$ contribution and arithmetic fluctuations. The formula permits large exact experiments without polynomial factorization. Such experiments should precede any proposed secondary constant; this article does not conjecture an unevidenced one.

8. Allow two independent generators. For multiplicatively independent positive algebraic numbers r, s , study solutions to $x + y = 1$ in $\{\pm r^a s^b : a, b \in \mathbb{Z}\}$. General unit-equation theory gives the surrounding framework, but a sharp, explicit analogue of the cyclic classification would require additional arithmetic input. Begin with fields and generators already appearing in the manuscript, and state the equivalence under a change of generators carefully.

8.3 Uniform asymptotics and gamma geometry

9. Connect the proportional and critical harmonic regimes. Construct an error estimate uniform as $\alpha = p/h$ grows from a positive constant to $\log h + O(1)$. The present rate satisfies $I(\alpha) \sim e^{-\alpha}/2$, while the preceding report has a critical limit $\exp(-e^{-c}/2)$ for $\alpha = \log h + c$. These formulas match formally, but a connecting theorem needs moving saddles, parameter-dependent curvature, and quantitative tail bounds. It should recover both the present $D(\alpha, a)$ correction and the earlier critical correction in their overlapping range.

10. Let both reflected gamma indices grow. For $m/n \rightarrow \theta > 0$, the integrand can be written

$$\exp\{n[\log(\log \Gamma(x)) + \theta \log(\log \Gamma(1-x))]\}.$$

The endpoint mechanisms of the fixed- m theorem then change. Locate the dominant point or points, establish uniqueness or a precise bifurcation statement, and derive a uniform Laplace expansion. At $\theta = 1$, symmetry forces the phase to be symmetric about $1/2$; it does not by itself prove that $1/2$ is the global maximum.

11. Determine the optimally truncated signed remainder. The unsigned bound is sufficient for certified approximation, but it does not identify the leading exponentially small remainder when K lies near the least term. Develop a contour or inverse-function description that determines its sign, scale, and dependence on the fractional part of the optimal truncation index. The already proved late-coefficient law supplies the candidate least-term location; turning it into a remainder theorem is a separate analytic step.

12. Classify early residue zeros and sign changes. The first coefficient is positive, and the late sign is $(-1)^{k+m-1}$. Early behavior depends on m : even the second nonzero coefficient changes sign between $m = 1$ and $m = 2$. Determine whether any early coefficient vanishes and obtain explicit bounds for the onset of the eventual sign law as a function of m . An exact identity in the formal zeta-gamma polynomial algebra should be distinguished from a numerical zero after substituting Euler's constant and zeta values.

13. Develop weighted inverse-gamma moments. Replace $B(x)^m$ by an analytic weight $W(x)$ with a specified order of vanishing at zero. The residue calculation suggests

$$A_k(W) = \frac{1}{k} [x^{k-1}] \Gamma(1+x)^k W(x),$$

and the same Cauchy-circle method should yield a late-coefficient expansion when the transformed amplitude does not vanish at the critical point. Precisely identify the hypotheses under which this holds, including endpoint integrability, zeros at the saddle, and weights with their own singularities. Reflected integer powers are the fully proved case in this article.

These questions have different proof requirements. Exact word identities and algebraic searches are natural targets for small certificates. Uniform asymptotics require analytic remainder estimates. Statements about numerical period independence require arithmetic ideas beyond the reduction and numerical methods used here.

9 Reproducibility and proof dependencies

The archive contains the complete $\text{\LaTeX}$ source, the compiled PDF, proof certificates, verification programs, result records, the plotted figure and its source program, and the proposed integration edits. The programs are included so that all finite claims can be replayed independently of the discovery process. The analytic proofs are in the article; the numerical files are supplementary diagnostics.

9.1 Exact checks

The S_4 verifier requires only the Python standard library. It rebuilds the relation rows from the stored descriptors and verifies the target coefficient by coefficient using integers and rational fractions. Its recorded result is

convergent double-shuffle rows	713
single-divergence double-shuffle rows	181
lifted convergent distribution rows	17
convergent duality seeds	1
nonzero residual coefficients	0

The coefficient file is approximately 538 kB. A separate discovery program rebuilds a certificate by exact fraction-free elimination; it is optional for verification. Replaying the supplied certificate does not require it, a saved elimination matrix, or binary caches. The discovery run also succeeded and the newly generated certificate passed the independent verifier.

The ladder program completed 2,012 checks. Of these, 2,008 are exact algebraic or arithmetic checks, and four are high-precision numerical audits. They include 435 exact irreducible-quotient checks for $1 \leq a < b \leq 30$, the six predicted scaled-plastic collisions in that range, all specialized Rogers and Kummer arguments, the rational row combinations, and the counting and Möbius formulas through degree 100. The finite survey corroborates the classification; the all-exponent proof depends on the cited irreducibility theorem and the argument in the text.

The reflected-moment program completed 119 exact symbolic assertions. They check the first three residue-coefficient formulas, the finite reflection sums through $k = 10$, the Appell rules through degree ten, and the stated generating-function coefficients. Neither these checks nor the numerical asymptotic tests replace the continuation, saddle, or remainder proofs.

9.2 Numerical audits

At 70 working decimal digits, independent quadratures for the two sides of the S_4 identity differ by approximately 4.58×10^{-71} . The four plastic identities were evaluated at 120 working digits; the largest residual was below 8×10^{-121} . These figures describe the observed arithmetic of the stated runs. They are not rigorous interval enclosures.

The proportional-depth program evaluates 24 positive integrals at 70 digits, using $h = 20, 40, 80, 160$, $\alpha = 1/4, 1, 3$, and $a = 1/2, 6/5$. It checks the explicit rate derivative separately and

records both the leading approximation and the first correction. The corrected relative errors have the predicted h^{-2} scale in these experiments. The figure displays the exact rate and prefactor formulas, sampled numerically for illustration.

At 120 digits, the reflected-moment diagnostics compare 16 late coefficients, 9 finite-moment approximations and explicit bounds, and 3 independent direct- x versus transformed-variable quadratures. The late-coefficient tests include the predicted β_m term. The moment tests verify that the observed errors lie below the displayed bounds in the selected cases. The proof establishes those bounds for the entire stated range, not just these samples.

9.3 Reproduction commands

From the root of the extracted package, the exact Gaussian proof can be checked with

```
python code/verify_s4.py results/S4_certificate.json
```

The remaining supplied checks use `sympy`, `mpmath`, and, for figure regeneration only, `matplotlib`. The command

```
python code/run_checks.py
```

runs the packaged verifiers and writes a combined receipt. To rebuild the article use

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  polylogarithms_rigidity.tex
```

The optional full Gaussian discovery command is

```
python code/discover_s4.py results/S4_certificate_rebuilt.json
```

It does not overwrite the supplied proof certificate. The README documents individual commands, expected outputs, and the exact scope of the proposed patch. A manifest records the audited source revision and SHA-256 digests of the delivered files.

9.4 What has been certified

There are two mathematical proof mechanisms here: proofs from stated classical theorems and explicit analytic arguments, and a finite computer-assisted rational proof whose relation schemas are justified in the text. The article is not a Lean, Isabelle, or Coq development. No numerical lower bound on a period-space dimension, no general maximal ladder weight, and no historical priority over all equivalent formulas is asserted. The verified result is the collection of exact identities, classifications, asymptotic expansions, and error bounds with the hypotheses stated in their respective theorems.

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